

Madhavan Balaji

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TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Frontend & Backend:** FastAPI (Async), RESTful APIs, WebSockets, React, HTML, CSS, JavaScript, Data Structures & Algorithms
- **Databases, Cloud & DevOps:** PostgreSQL, asyncpg, MySQL, Docker, Kubernetes, ArgoCD, CI/CD (GitHub Actions), Slurm, GCP, AWS, Render Cloud, Git
- **Applied AI & ML:** LangGraph, CrewAI, LangChain, LlamaIndex, OpenAI APIs, HuggingFace Transformers, PEFT/LoRA, Model Observability, Core Evaluation, PyTorch, TensorFlow, Scikit-learn, Vision Transformers (ViT), Stable Diffusion, Qdrant, ChromaDB, FAISS, Sentence Transformers, Text Classification, Entity Extraction, Semantic Search

PROFESSIONAL EXPERIENCE

Technical Instructor & Curriculum Lead

Indiana University Indianapolis

Aug 2025 – Present

Indianapolis, IN

- Elevated academic performance and practical code implementation velocity across a **30+ student** cohort by architecting and delivering modular weekly technical workshops on core Data Structures, Algorithms, and optimization concepts through live-coding challenges.
- Boosted student programmatic engagement and performance benchmarking transparency by engineering a customized competitive programming leaderboard infrastructure tracking active weekly and cumulative overall class rankings.
- Enhanced code optimization capabilities and syntax architecture standard compliance by coordinating structured peer-review labs evaluating execution efficiency, complex algorithmic edge cases, and Big-O space-time complexity trade-offs.

Data Analyst Intern

Coincent

Sep 2022 – Nov 2022

Bangalore, India

- Achieved a **90% classification accuracy** threshold and enabled sub-second item query times by engineering a **CNN-based** fashion retrieval pipeline translating unstructured image streams into a searchable index matrix.
- Cut model feature extraction latency parameters by **30%** to unlock real-time streaming pipeline throughput by profiling bottlenecks and implementing localized structural image preprocessing optimization loops.
- Generated a measured **40% lift** in simulated user engagement analytics metrics by developing and training a robust multi-class classifier spanning **1,000+ distinct clothing categories** to power downstream recommendation engines.

TECHNICAL PROJECTS

Sovereign SRE – Autonomous Self-Healing Infrastructure | [GitHub](#) | [Live App](#)

May 2026

- Automated 100% of initial production incident handling steps by designing an autonomous self-healing SRE architecture pairing a stateful **LangGraph** routing engine with a **CrewAI** multi-agent collective for root-cause analysis and patch generation.
- Eliminated manual code response risks by integrating localized **Ollama (Llama 3.3)**, **Anthropic**, and **Gemini** models secured by programmatic Human-in-the-Loop validation gates and branch-verified **GitHub** Pull Request generation.
- Established real-time system observability by orchestrating decoupled agent microservices via **Docker Compose** utilizing asynchronous **WebSocket** data streams to supply telemetry to a **Next.js** interface hosted on **AWS Amplify**.

DocuMind – Multi-Agent Research Intelligence Platform | [GitHub](#) | [Live Demo](#)

Jan 2026

- Optimized asynchronous document extraction pipelines to achieve an operational throughput of **167 texts/sec** and **sub-15ms** p50 latency by deploying a stateful **LangGraph** hybrid RAG system with dense **Qdrant** search and **CrossEncoder** reranking.
- Guaranteed absolute data durability and multi-agent state survival under high concurrency loads by architecting a non-blocking **asyncpg/PostgreSQL** persistence layer tied to a background worker pool and a live telemetry interface.
- Removed workflow context-switching penalties by implementing and open-sourcing a custom Model Context Protocol (**MCP**) server registering **6 highly functional tools** for immediate native integration inside Claude Desktop clients.

DentaVision – Vision Transformer SaaS | [GitHub](#) | [Live App](#)

May 2025

- Elevated clinical detection accuracy parameters to **92.4% top-1 accuracy** (+8.1 pts over baseline models) by fine-tuning an open-weight **Vision Transformer (ViT)** served via an optimized **FastAPI** inference layer running at a P95 threshold of 310ms.
- Accelerated report generation speeds by **41%**, lowering time-to-first-prediction down to **1.2s**, by developing a responsive **React 18** and **TypeScript** interface translating model logit matrices into clean diagnostic structures.
- Ensured cross-platform deployment environment consistency by containerizing backend and frontend layers within isolated **Docker** modules optimized with stateless operational logic for deployment on **Render**.

PATENT

Published: Caremate – An Integrated Elderly Support System | [View Document](#)

EDUCATION

Indiana University Indianapolis

Master of Science in Computer Science

May 2026

Indianapolis, IN

- **Relevant Coursework:** Deep Learning, Machine Learning, Computer Vision, Algorithms, Data Mining, Database Design, Natural Language Processing, Cloud Computing

Sri Venkateswara College of Engineering

Bachelor of Information Technology

2024

Chennai, India